Lesson notes · for you

Science stand-ins, and one machine

AI · University · Lesson 4

Materials
A phone with Darsi; Paper for one hand-worked step; The slides from the kit

Learning goals

- Fast where it was tested
- Not the experiment

Activity: In the lab

1. Choose “A wide layer”. Try “Split the examples” (32, more than 28), then “Split the layer” (24, it fits).
2. Choose “A large batch” and “Split the examples”: 4 + 4 × 4 = 20, which fits in 22. Write which split fitted each job.

Discussion

- What does a bigger machine not answer?

Check yourself

- Wide layer: which split fits?

Try it this week
Compute the wide layer’s memory both ways on paper and check the lab.

Worksheet

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Name _____ Class _____ Date _____

1 **Is a fast stand-in for a slow experiment the experiment itself?**

- ◯ Yes, for any case once it is trained
- ◯ Yes, if it was accurate on its tests
- ◯ No: it is trusted only where it was tested

2 **Four machines that hold 28 each share the wide layer: 16 weights and a batch of 2. Split the layer four ways. How much memory does each machine need?**

3 **Match the split.**

Data parallel	• • • •	Each machine holds part of the layer
Model parallel	• • • •	One machine, so this lab is a diagram
One phone	• • • •	Each machine holds the whole model and some examples

4 **Order a decision about a wide layer.**

- ☐ Try splitting the examples
- ☐ Split the layer if that still does not fit
- ☐ Write the memory of one step

✎ **When does splitting the examples fail?**

Check your answers

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1 **No: it is trusted only where it was tested**

A stand-in answers fast where it was tested.

2 **12**

Each machine keeps a quarter of the weights, 4, and their activations for both examples, 2 × 4: 4 + 8 = 12.

3 **Data parallel — Each machine holds the whole model and some examples**
Model parallel — Each machine holds part of the layer
One phone — One machine, so this lab is a diagram

A step stores weights, gradients and optimiser state plus activations, and activations grow with the batch.

4 **Top to bottom: 2, 3, 1**

You measure before you name a bigger machine.